RCA, Outage, and Licensing Projects

Notes about common architecture elements

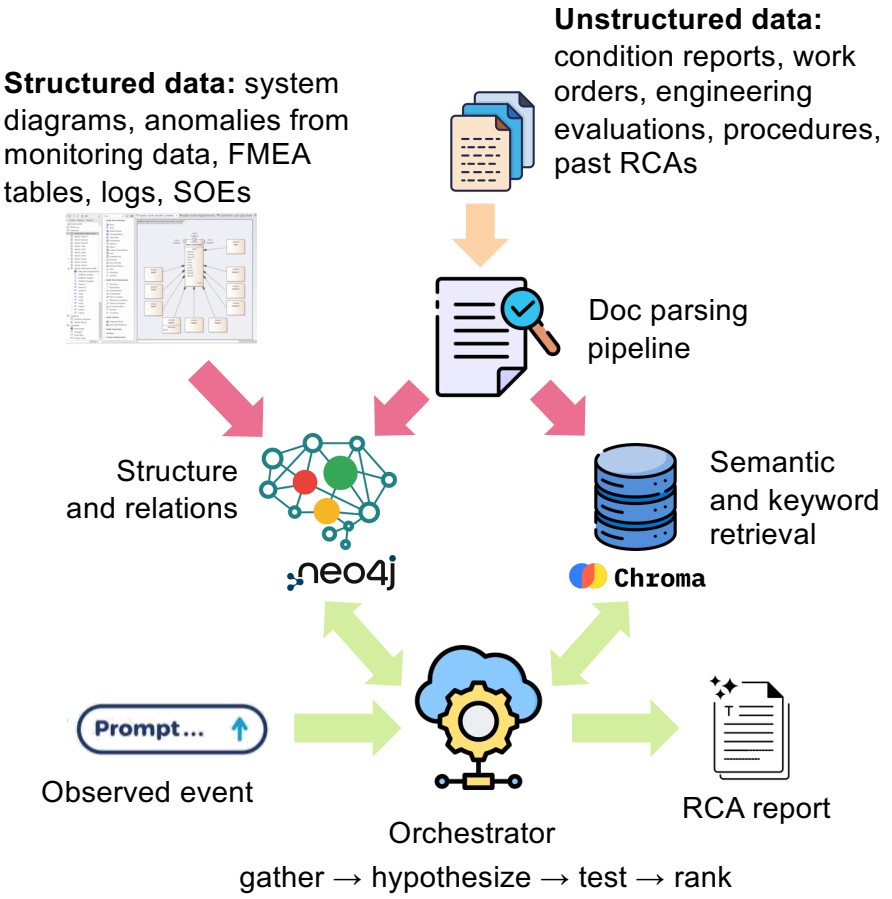
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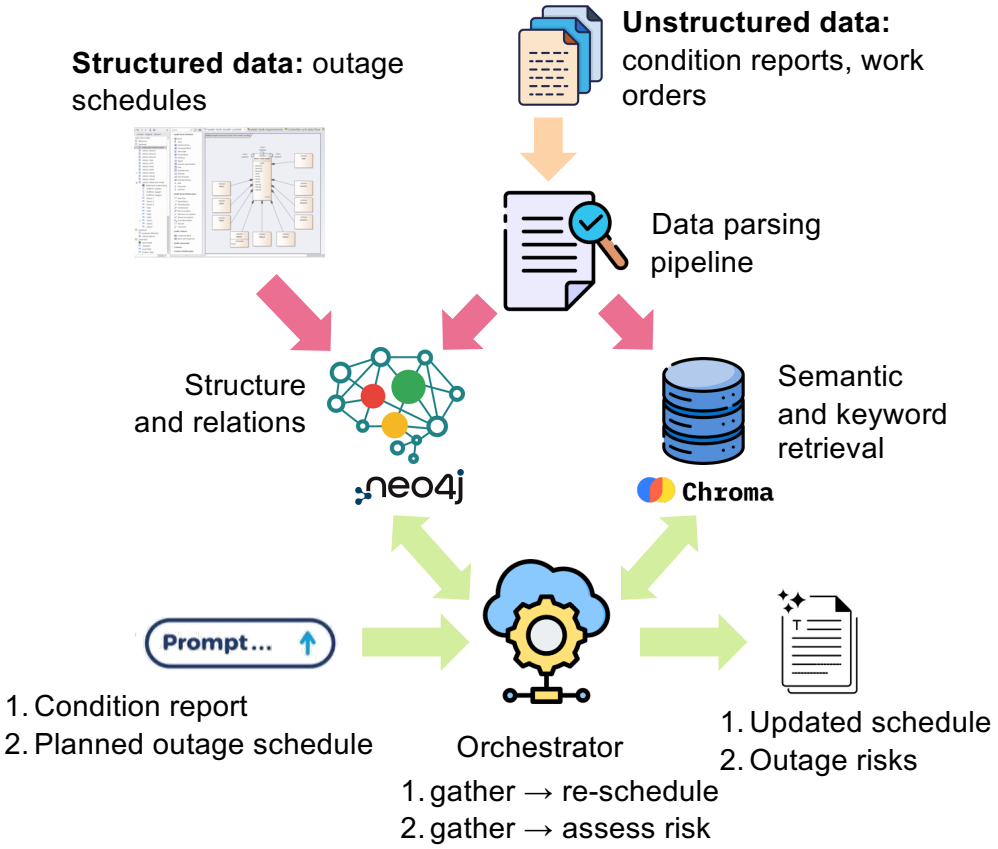
RCA Project

- **Goal:** Co-pilot engineers to improve root cause analysis (RCA) for complex events by combining plant data, and system knowledge into a unified reasoning workflow
- **Pipelines**
  A. Parse and extract (text → knowledge)
  B. Populate data structures (Chroma and knowledge graph)
  C. Analyze (reason over the built structures)
     1. Structural reasoning
     2. Temporal reasoning
     3. Evidence-based reasoning
- **Data analytics methods**
  - Knowledge extraction + epistemic
  - Temporal analysis and pattern recognition methods
  - PM compliance + cross-source corroboration
  - Causality engine



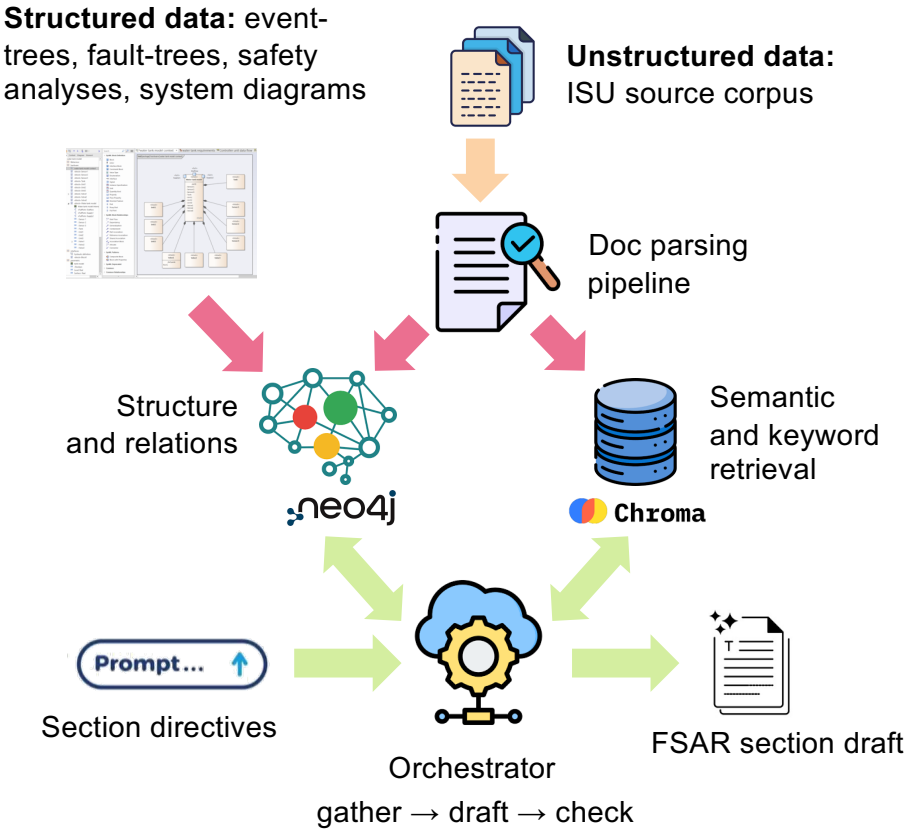
Outage Project (FY-27)

- **Goal:** Provide decision support to outage managers and staff during outage planning and execution phases
  - Reduce workload on outage managers and plant staff throughout outage lifecycle
  - Reduce risk of outage delays cause by delays and unplanned activities
- **Pipelines**
  - **Parse and extract:** Historical condition reports, work orders, outage schedules
  - **Assess delay:** Assess activity completion uncertainty
  - **Assess emergent work**: Assess risk of un-planned activities
- **Data analytics methods**
  - Similarity search
  - Knowledge extraction
  - Resource-constrained outage scheduler
  - Schedule optimization methods



Licensing Project (ISU)

- **Goal:** An AI-assisted workflow for drafting FSAR sections according to licensing directives (e.g., based on NUREG-1537)
- **Data corpus:** Hundreds of documents across PDF, DOCX, XLSX, superseded and current versions
- **Pipelines**
  A. Parse and extract (text → knowledge)
  B. Populate data structures (Chroma and knowledge graph)
  C. Draft & review: Retrieve, assemble evidence package, generate cited text, run guardrails



KG Schemas

- **Major classes**
  - **MBSE:** Derived from UML (mostly) and Sysml (ports)
    - System decomposition: plant → system → subsystem → equipment → component
    - Functional relations
  - **Safety risk**
    - PRA, FMEA, HAZOP, STPA
  - **Operations**
    - CR, WO, outage, equipment monitoring, equipment operations, numeric performance, supply chain
  - **Analysis**
    - Simulation, req.s and tech. specs, RCA
  - **Generic**
    - Document, temporal relations, codes/standards/regulatory docs
- **To be addressed**
  - Set type vs. instance
  - MBSE spatial decomposition
  - Temporal modeling
    - Phases: design, construction, commissioning, operation, maintenance, decommissioning
    - Support states/configurations of the same entity over time (as-designed, as-built, as-operated, as-decommissioned)

Observations

- These projects rely on a set of common data structures
  - **Knowledge graph:** source of truth for any data element
    - Based on a set of defined schemas
    - Should contain only the metadata of a data element (agree?)
  - **Chroma:** Semantic/keyword search over text chunks
  - **LLM Wiki:** Maintained interpretive pages (per system / per equipment/ per failure mode / per recurring anomaly class, etc.) organized around well defined KG anchors
- Each project has a set of data analytics engines that act upon these data structures
  - Developed pipelines are fairly hard-wired
  - Ideally, they would act upon these data structures using a common protocol (use case agnostic)

Scope of this slide set

- Gather ideas on
  - How the set of data structures (contained in DACKAR) can be made agnostic of the project (or use case)
  - Define APIs for the data analytics methods to act on these data structures
- To be discussed
  - How to build robust data ingestion pipelines
  - How to integrate data structures KG, Chroma, and LLM wiki
    - KG for structured data via deterministic ingestion
    - Chroma for unstructured data (semantic and keyword retrieval)
    - LLM wiki for unstructured data
      - e.g., page for per subsystem, per failure mode, per recurring anomaly class
      - LLM-based and deterministic ingestion
  - How to formalize APIs for analytics engines (either agents of deterministic methods)
    - Read/write protocols
      - Including provenance
- Are elements in Prometheus (e.g., deeplinx) that can be leveraged?
  - How to interface DACKAR and LOGOS methods?

October 12th Meeting

- [M]
  - one ingestion pipeline: merge with DACKAR (check for conflict)
    - [C] Check for config file
    - [C] Add LLM wiki
  - Add verification piece for checking
  - Issues with handling of typos or abbreviations or case-sensitive search
    - [C] Use graph database and semantic search
  - Use LLM to query databases (SQL and [C] cypher)
  - Issues to be discussed
    - When we decide on how to use internet-retrieved material? Develop rules!
      - Provide valid sources for internet retrieved data elements
    - Issues with number and units
    - Inconsistencies on data generation from retrieved data (e.g., from tables)
    - Analysis/judgment on drafted analysis
    - Vague statements
    - Use decision LLM from jev (rely on https://github.com/NandhaKishorM/laya?)
    - SLIDERS: https://arxiv.org/abs/2604.22294
- [C]
  - Restructure of DACKAR and check repo (e.g., out of RCA folder)
  - Decide on how to design API for data structures
    - Set discussion meeting (meeting set)
    - Decide on level of interaction (both agent and human in the loop)
    - Check with Prometheus development
  - Platform to use? Streamlit? Langchain? Protocol to use?
    - [M] Json should be enough
  - Draft tasks for all projects
  - Rely more on decision models to assess/rank issues
    - Outage might be a simple setting to test it (e.g., similarity search for outage, temporal pattern)
    - How to measure confidence that is actually meaningful

Action items

1. [All]: Draft notes and observations about APIs for data structures to be discussed on the Oct 27 meeting
   - Query through natural language on multiple data structures?
2. ~~[D]: complete merging of RCA work~~
3. [D]: complete review and merging of outage work
4. [C+M]: LLM wiki development
5. [C+D]: update and check DACKAR repository
6. ~~[D]: draft tasks for all 3 projects~~ (see slide 10)
7. [C+M]: Merge portions of ISU github repo in Dackar (check first element 9)
   - merge latest development of branches of ISU github repo in ”main”
   - Put on hold to wait for Mengnan agentic work demonstration
8. [All]: define and build data ingestion pipeline
   - Might require on item 1., 4. and 7.
9. [D]: Draft Licensing agent document to be reviewed

Projects Tasks

RCA	Outage	Licensing
• Analyze OPG data • Gather more data from OPG • Analyze Xcel data • Draft agentic infrastructure • Develop GUI	• Finalize activity similarity search • Develop handler for unplanned activities (outage planning phase) • Develop handler for delays, unplanned activities, and resource changes (outage execution phase) • Integrate DACKAR with LOGOS • Develop test and demonstration cases • Draft agentic infrastructure • Develop GUI	• Finalize ingestion pipeline • Draft agentic infrastructure